

Midterm-I Exam

Total Time (min): 1 hr

Total Marks: 30

Total Questions: 3

Applied Human Computer Interaction (CS 3014)

Date: 24th Feb, 2026

Course Instructor(s)

Mr. Kasahn Hussain, Mr. Shaheer Ahmad, Mr. Ghulam Murtaza, Mr. Asif Fazwani and Mr. Jibran Rasheed

Roll No

Section

Student Signature

Do not write below this line.

Attempt all the questions. Do the questions in order, out of order questions will not be marked

CLO # 1: Explain fundamental concepts, theories, and models of Human Computer Interaction

Q1a) A software engineer has taken the requirements from a corporate customer, the requirement consists of some paragraphs which need thorough reading to understand the intention correctly. He proceeds with the project and after one month of duration completes the project and handed over successfully to the client in time. Before handing over he performed final quality test of his product in which he found some faults due to incorrect perception building and thorough understanding of the requirements. Which point/s and steps regarding reading the requirement he missed? Support your answer with appropriate HCI concepts only **[05 points]**

Q 1b) A computer engineer in an XYZ company starts a project of vehicle tracking, in the meanwhile he assigned few other projects from the Project Manager. After a considerable amount of time, the Project Manager checks the progress and indicate the shortcomings from the vehicle tracking project and mentions that the engineer could not remember the project's minimum requirement in essence, explain the reasons of forgetting in the context of HCI concepts only. **[05 points]**

CLO #1: Explain fundamental concepts, theories, and models of Human Computer Interaction

Q.2a) An HCI researcher observes user performance while interacting with a menu-driven interface.



Out of 100 users:

[5 points]

- 80 users successfully completed the task using Menu A.
- 20 users failed to complete the task using Menu A.

Later, a new user attempts the same task using Menu A.

(i) What type of reasoning and conclusion might the researcher draw about the success of Menu A?

(ii) The system design rule states:

“If task success rate $\geq 70\%$, the interface is considered usable.”

Show the reasoning type and also determine whether Menu A satisfies the design rule.

Show your reasoning steps.

Q2b) Most Automated Teller Machines (ATMs) return the debit card before dispensing cash. Why?

In your answer, discuss how human cognition, perception, and user behavior influence this design choice.

Support your explanation with appropriate HCI concepts.

[5 points]

CLO # 1: Explain fundamental concepts, theories, and models of Human Computer Interaction

Q3a) A spy intercepts two short coded messages while hacking a secure communication network of a government organization as shown below:

- **Message 1:** A B C D E
- **Message 2:** A A B A A

Using Shannon Entropy method explain which message is of more importance to the spy with respect to information and uncertainty?

[5 points]

Q3b) Sarah wants to listen to music on her tablet. She opens the music app and taps the "Play" button on her favorite album. The screen flashes brightly for a second, but no music plays.

Considering the above scenario briefly explain the complete Donal Norman model of interaction.

Remember to follow the sequence of model.

[5 points]